

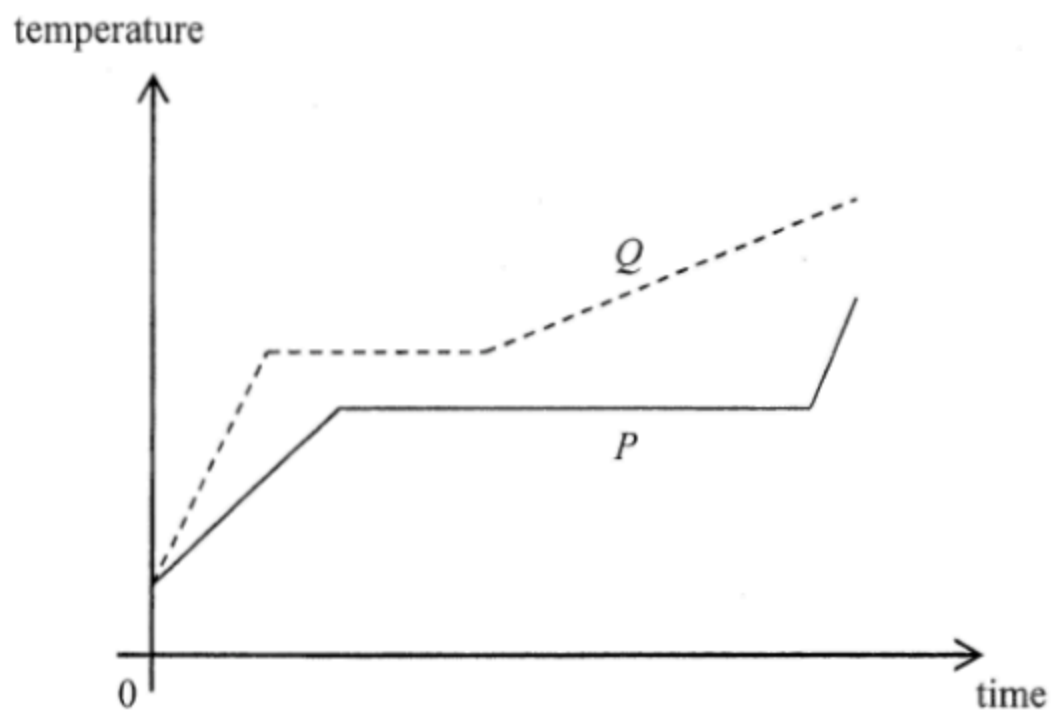
1. 30 g of milk at  $10^{\circ}\text{C}$  is added to 120 g of coffee at  $80^{\circ}\text{C}$ . Assuming there is no heat loss to the surroundings, what is the final temperature of the mixture?

Given: specific heat capacity of milk =  $3800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

specific heat capacity of coffee =  $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

- A.  $64.8^{\circ}\text{C}$
- B.  $65.2^{\circ}\text{C}$
- C.  $66.0^{\circ}\text{C}$
- D.  $67.1^{\circ}\text{C}$

2. Same mass of solids  $P$  and  $Q$  are heated at the same rate. The temperature-time graphs of the two substances are shown below.



Which of the following comparisons about their melting points and specific latent heats of fusion is correct ?

|    | higher melting point | larger specific latent heat of fusion |
|----|----------------------|---------------------------------------|
| A. | $P$                  | $P$                                   |
| B. | $P$                  | $Q$                                   |
| C. | $Q$                  | $P$                                   |
| D. | $Q$                  | $Q$                                   |

3. Which of the following statements about the internal energy of a substance are correct ?

- (1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
- (2) When a vapour condenses, its internal energy decreases.
- (3) When a liquid evaporates, the internal energy of the remaining liquid increases.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

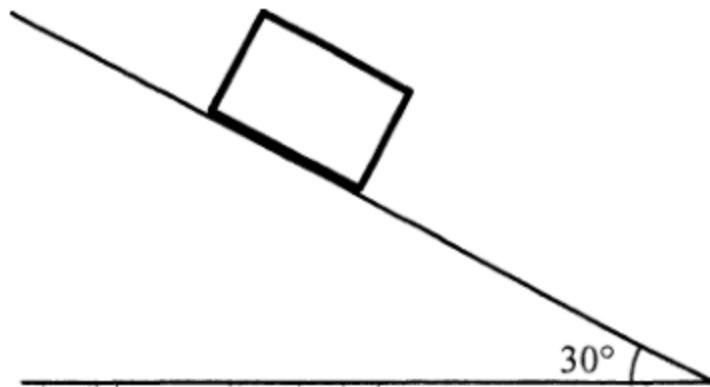
\*4. The pressure of a fixed mass of an ideal gas at  $10^{\circ}\text{C}$  is  $2 \times 10^5 \text{ N m}^{-2}$ . If the volume of the gas is reduced to half of its original volume and its temperature is increased to  $100^{\circ}\text{C}$ , what would the pressure be ?

- A.  $1.00 \times 10^5 \text{ N m}^{-2}$
- B.  $1.32 \times 10^5 \text{ N m}^{-2}$
- C.  $4.00 \times 10^5 \text{ N m}^{-2}$
- D.  $5.27 \times 10^5 \text{ N m}^{-2}$

5. Which of the following statements about the motion of any two objects is correct ?

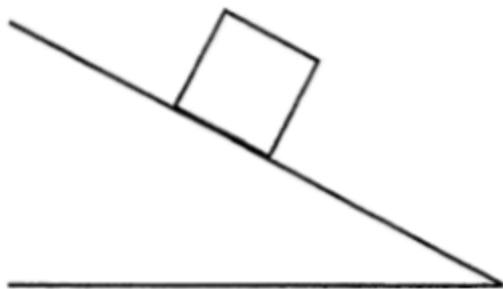
- A. The object that takes a shorter time to complete the same path must have greater average speed.
- B. The object that travels a greater distance in 1 s must have greater average velocity.
- C. The object with greater velocity must have greater acceleration.
- D. If the two objects have the same acceleration, they must be moving in the same direction.

6. A block is released from rest on an inclined plane as shown. The inclined plane makes an angle of  $30^\circ$  to the horizontal. The block moves with uniform acceleration, and travels a distance of 1 m in the first 3 s. Determine the acceleration of the block.



- A.  $0.22 \text{ m s}^{-2}$
- B.  $0.33 \text{ m s}^{-2}$
- C.  $4.91 \text{ m s}^{-2}$
- D. Cannot be determined as the frictional force acting on the block is unknown.

7. A block of mass  $m$  stays at rest on a rough inclined plane as shown.

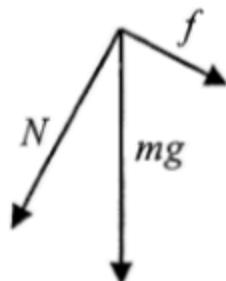


Which of the following diagrams correctly shows the forces acting on the block ?  
( $N$  is the normal reaction from the inclined plane, and  $f$  is the frictional force between the block and the plane.)

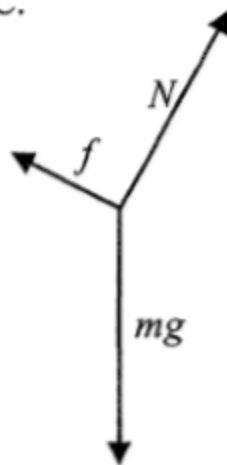
A.



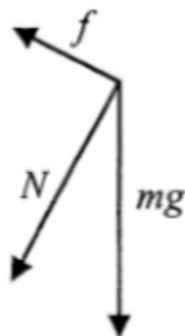
B.



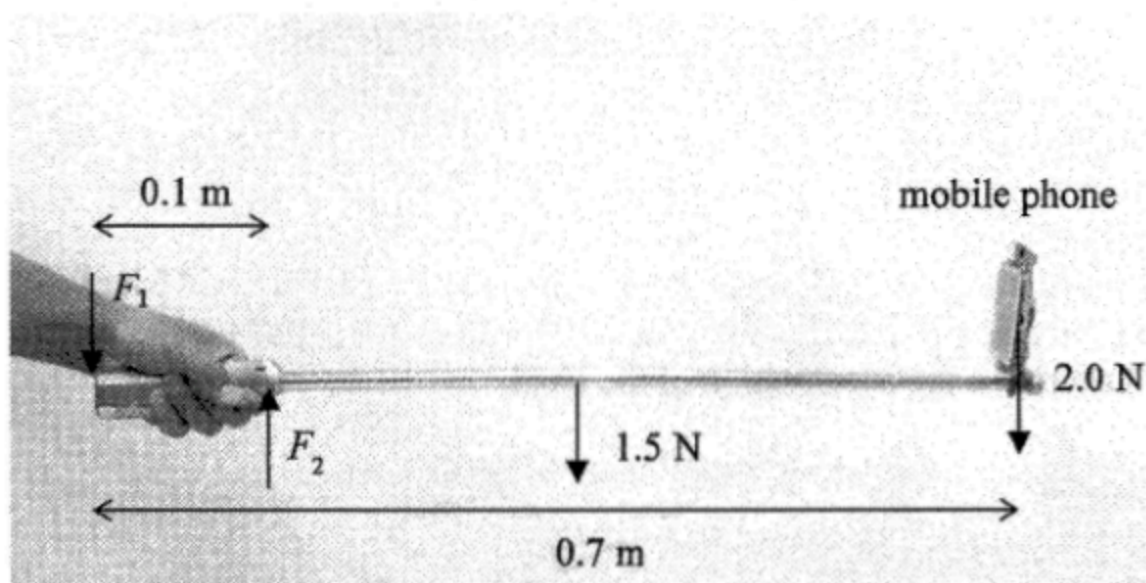
C.



D.



8. Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown. Assume that the forces required to hold the selfie stick by the hand are represented by  $F_1$  and  $F_2$ , and  $F_1$  and  $F_2$  are perpendicular to the stick.

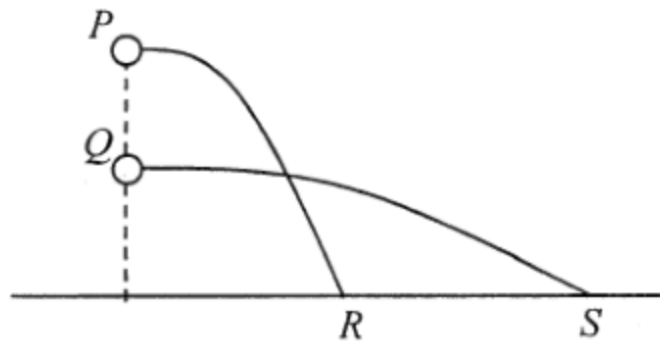


It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of  $F_2$ .

- A. 3.5 N
- B. 19.3 N
- C. 35 N
- D. Cannot be determined as  $F_1$  is unknown.



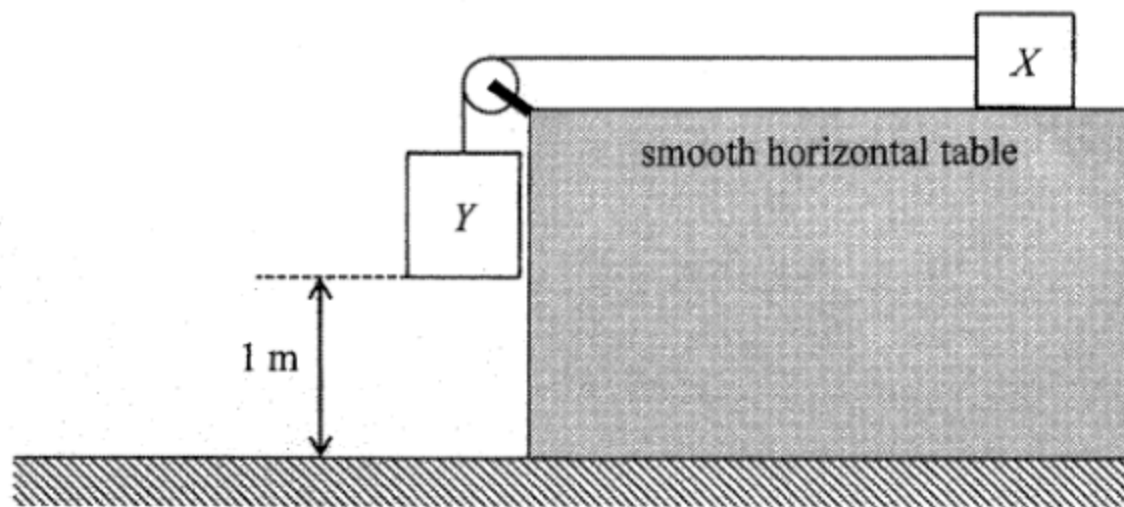
- \*9. Marbles  $P$  and  $Q$  of the same mass are shot horizontally. They hit the horizontal ground at points  $R$  and  $S$  respectively as shown. Neglect air resistance.



Which of the following statements is **INCORRECT** ?

- A. The initial speed of marble  $P$  is smaller than that of marble  $Q$ .
- B. The time of flight of marble  $P$  is shorter than that of marble  $Q$ .
- C. The potential energy loss of marble  $P$  is greater than that of marble  $Q$ .
- D. The acceleration of marbles  $P$  and  $Q$  is the same during the flight.

10. Blocks  $X$  and  $Y$  are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The mass of  $X$  and  $Y$  are  $0.5\text{ kg}$  and  $1\text{ kg}$  respectively. Initially,  $Y$  is  $1\text{ m}$  above the ground and the string is taut. The system is then released from rest.

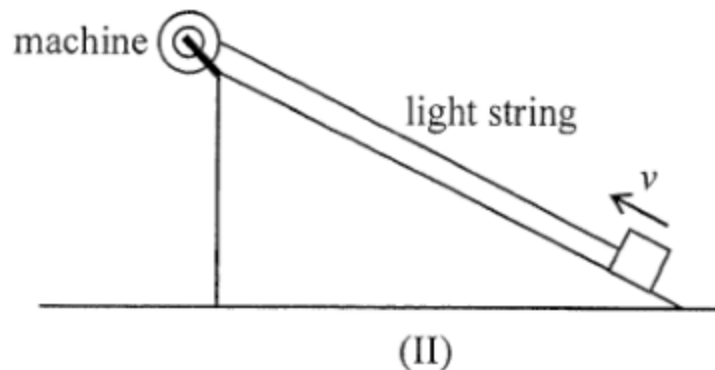
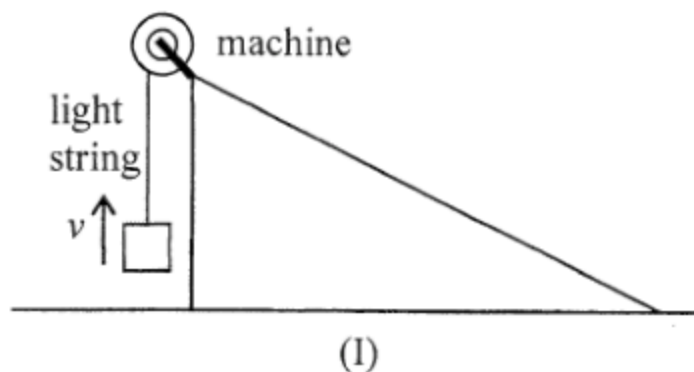


What is the speed of  $Y$  just before it reaches the ground? (Take  $g = 9.81\text{ m s}^{-2}$ )

- A.  $3.62\text{ m s}^{-1}$
- B.  $4.43\text{ m s}^{-1}$
- C.  $6.26\text{ m s}^{-1}$
- D.  $9.81\text{ m s}^{-1}$

11. A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

- (I) Pull the block vertically upward at a uniform speed  $v$ .  
(II) Pull the block up along the inclined plane at the same uniform speed  $v$ .

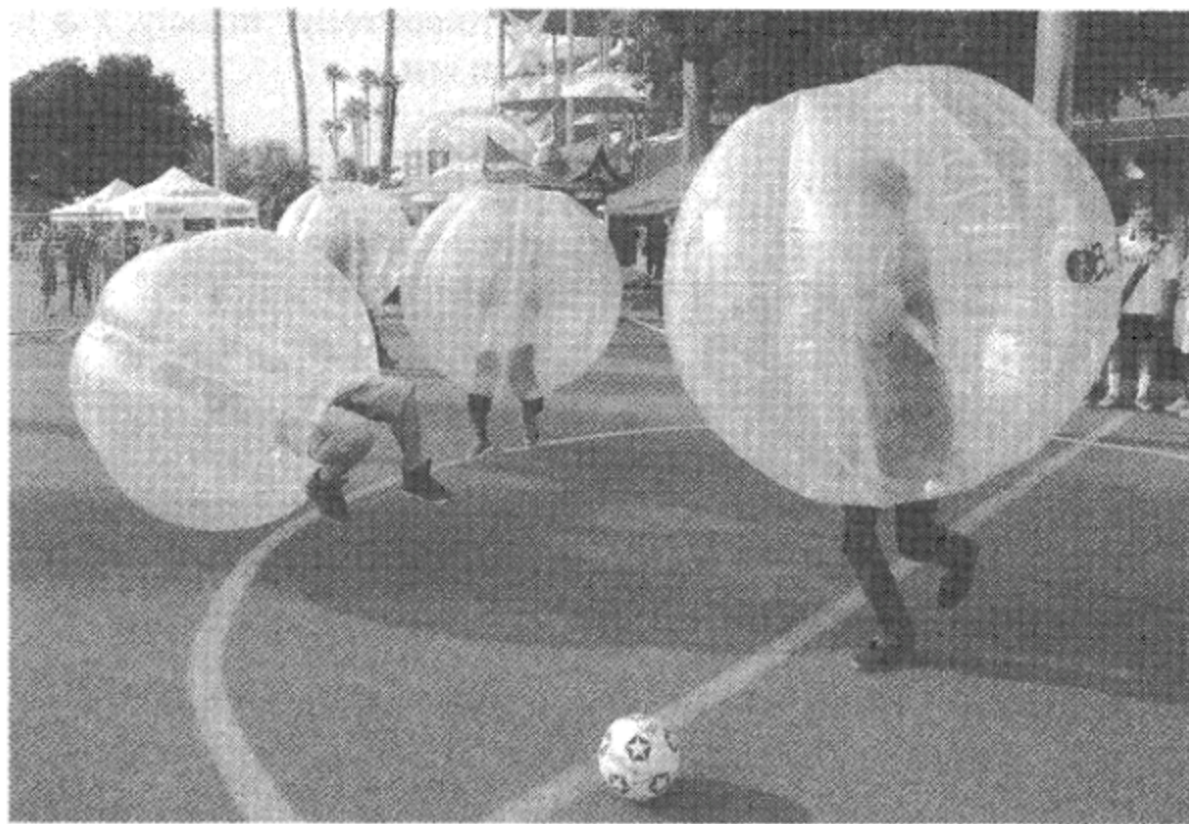


Which of the following statements correctly compare(s) the two methods ?

- (1) The tension in the string is the same.  
(2) The average output power of the machine is the same.  
(3) The work done by the machine on the block is the same.

- A. (1) only  
B. (3) only  
C. (1) and (2) only  
D. (2) and (3) only

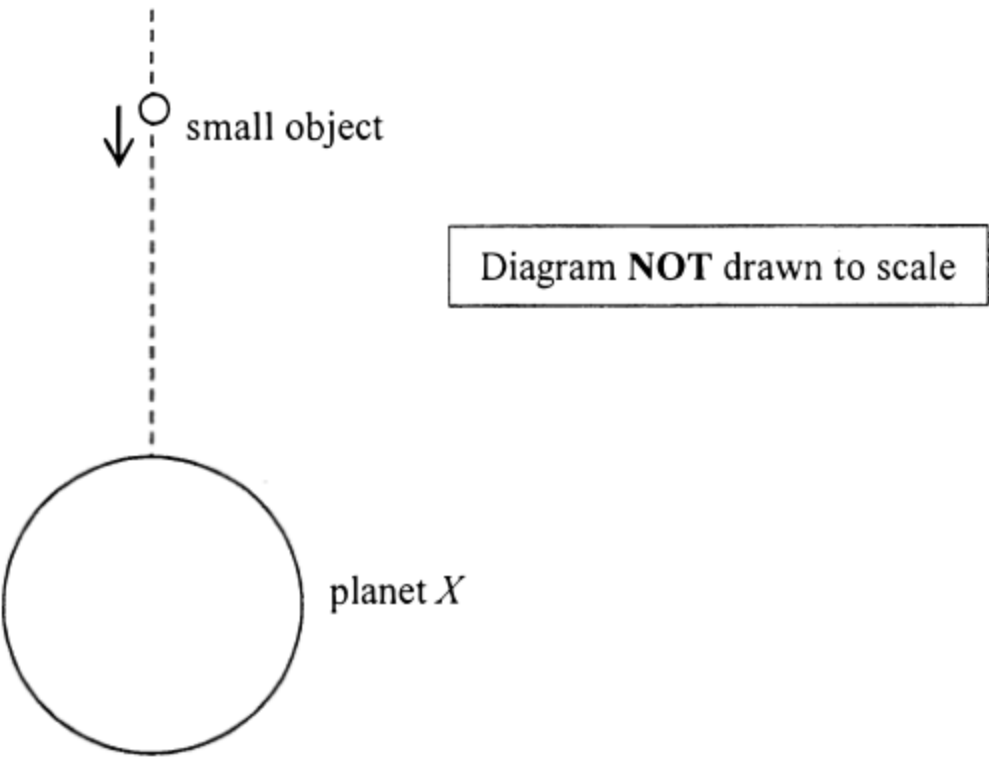
12. Players of "bubble soccer" wear air-filled plastic "bubbles" as shown.



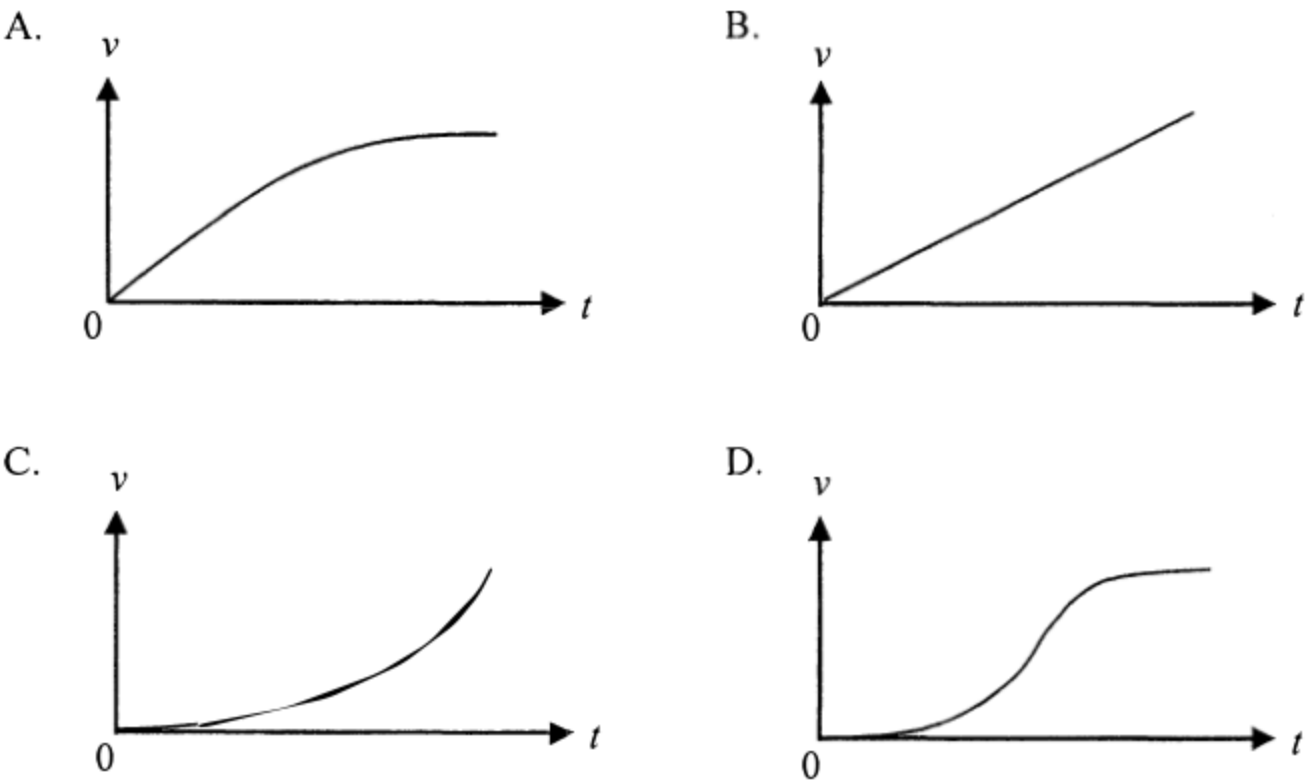
Which of the following statements best explains why the bubble can reduce the chance of injury during a collision ?

- A. The bubble increases the mass of the player, thus the momentum of the player increases.
- B. The bubble increases the air resistance acting on the player.
- C. The bubble lengthens the impact time during a collision.
- D. Like a balloon, the bubble provides a lifting force to the player.

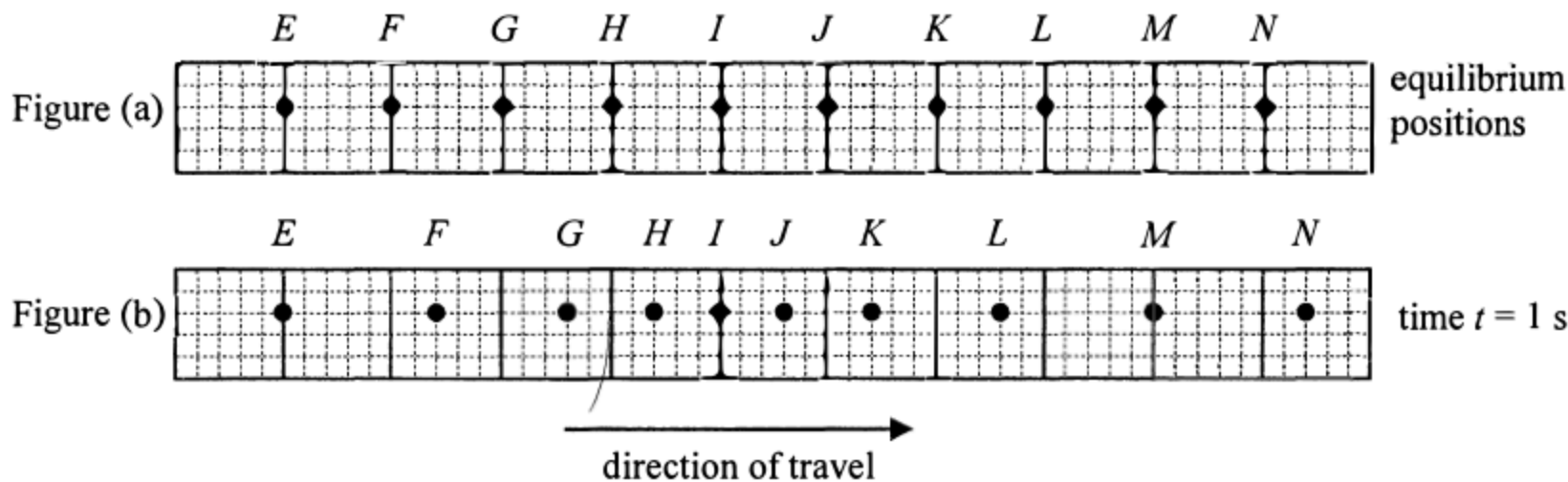
\*13. A small object is released from rest at a point very far away from a planet  $X$ . The object then starts moving towards  $X$ .  $X$  does not have an atmosphere. Neglect the effect of other celestial bodies.



Which of the following graphs best shows the variation of the velocity  $v$  of the object with time  $t$  before it hits  $X$ ?



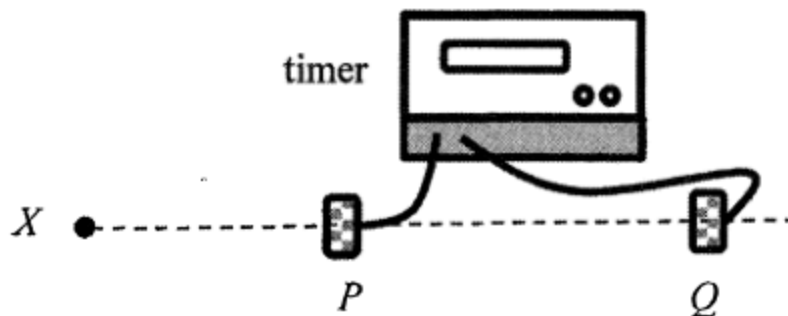
14. Figure (a) shows the equilibrium positions of particles  $E$  to  $N$  in a medium. At time  $t = 0$ , a longitudinal wave starts travelling from left to right. At time  $t = 1$  s, the positions of the particles are shown in Figure (b).



Which of the following statements **MUST BE** correct ?

- A. The distance between particles  $F$  and  $N$  is equal to the wavelength of the wave.
- B. The period of the wave is 1 s.
- C. Particle  $E$  is always at rest.
- D. Particle  $I$  is momentarily at rest at  $t = 1$  s.

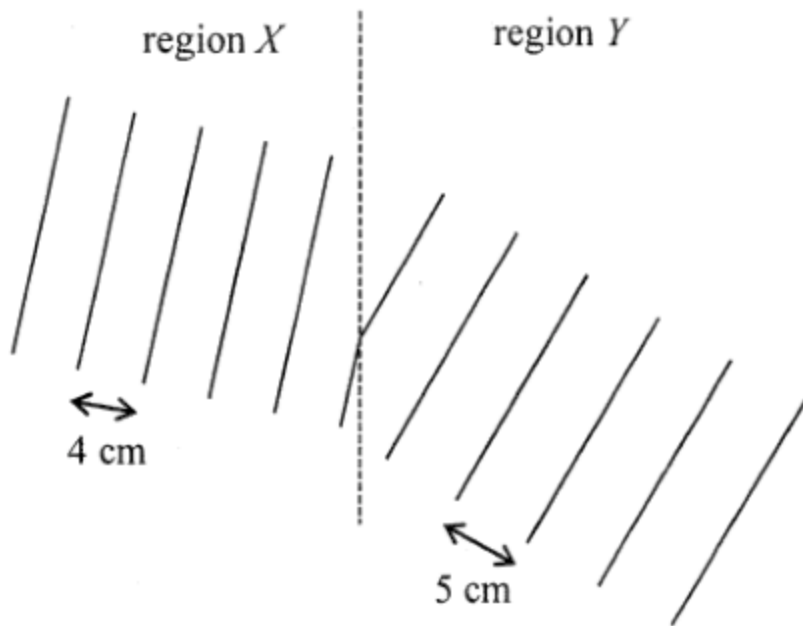
15. An experiment is set up to measure the speed of sound in air as shown.  $P$  and  $Q$  are two microphones connected to a timer. A sound is produced at  $X$ . The timer starts when  $P$  receives the sound, and stops when  $Q$  receives the sound. The timer shows the time taken for the sound to travel from  $P$  to  $Q$ . The distance  $PQ$  and the time shown can be used to calculate the speed of sound.



Which of the following statements is **INCORRECT** ?

- A.  $X$ ,  $P$  and  $Q$  must be along the same straight line.
- B. The percentage error in the time measured will increase if the distance  $PQ$  is reduced.
- C. The speed of sound determined should be independent of the distance between  $X$  and  $P$ .
- D. The distance  $PQ$  must be equal to an integral multiple of wavelengths of the sound produced at  $X$ .

16. The figure shows plane water waves travelling from region  $X$  to region  $Y$ . The wavelengths of the water waves in regions  $X$  and  $Y$  are 4 cm and 5 cm respectively.



Which of the following statements is correct ?

- A. The speed of the water waves in region  $X$  is higher than that in region  $Y$ .
- B. The direction of travel of the water waves bends towards the normal as they enter region  $Y$ .
- C. The frequency of the water waves is the same in both regions.
- D. If plane water waves of wavelength 5 cm travel from region  $Y$  to region  $X$ , the wavelength becomes 6 cm after the waves enter region  $X$ .



17. In which of the following situations **MUST** the direction of travel of a wave change ?

- (1) when a wave is reflected by a barrier
- (2) when a wave enters from one medium to another medium
- (3) when a wave travels through a gap smaller than its wavelength

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

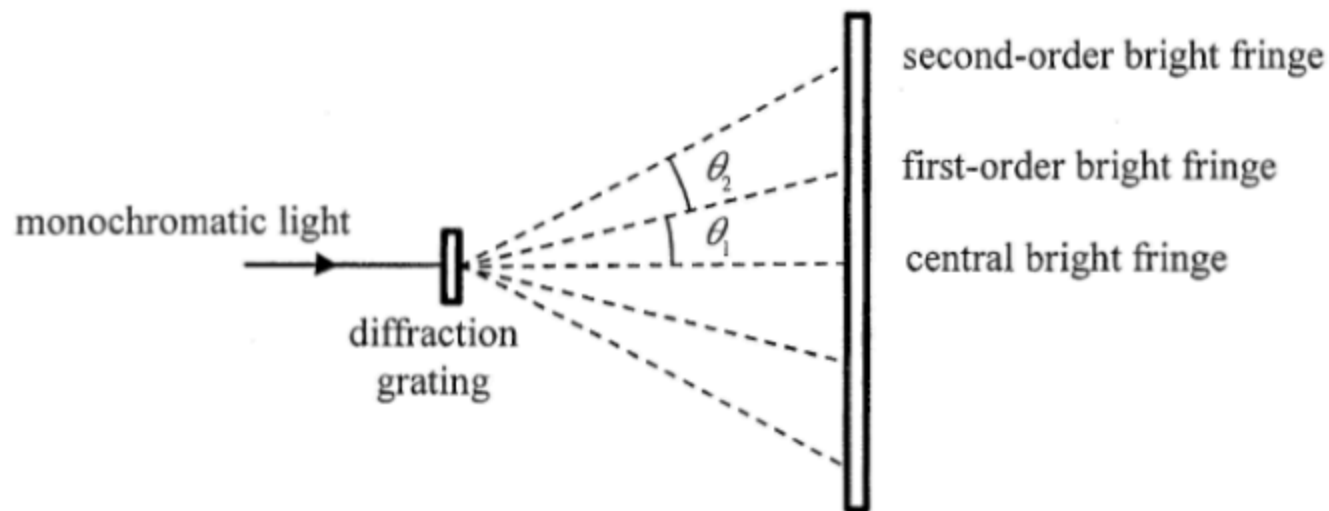
18. Two musical notes of the same pitch and loudness are produced by two different musical instruments. They sound different to the human ears because they have different

- A. amplitudes.
- B. phases.
- C. wave speeds.
- D. waveforms.

- \*19. When an object is placed 30 cm in front of a concave lens, an image is formed 20 cm away from the lens. If the concave lens is replaced by a convex lens of the same focal length and the object distance remains unchanged, which of the following descriptions about the image formed is correct ?

|    | <b>nature of the image</b> | <b>image distance</b> |
|----|----------------------------|-----------------------|
| A. | real                       | 20 cm                 |
| B. | real                       | 60 cm                 |
| C. | virtual                    | 20 cm                 |
| D. | virtual                    | 60 cm                 |

- \*20. The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.



Which of the following is/are correct ?

- (1)  $\theta_1 = \theta_2$
- (2) The maximum order of bright fringe is 4 if  $\theta_1 = 20^\circ$ .
- (3)  $\theta_1$  will decrease if the experiment is performed in water but not in air.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

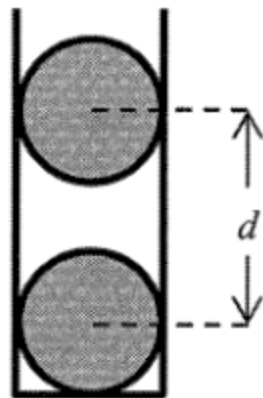
21. If the speed of sound in water is  $x$  and the speed of light in water is  $y$ , which of the following is correct ?

speed of sound in air

speed of light in air

- |    |       |       |
|----|-------|-------|
| A. | $> x$ | $> y$ |
| B. | $> x$ | $< y$ |
| C. | $< x$ | $> y$ |
| D. | $< x$ | $< y$ |

22. In the figure, two charged conducting spheres of the same mass  $m$  are put in a vertical plastic cylinder. The inner wall of the cylinder is smooth. The spheres are separated by a distance  $d$  and remain in equilibrium.

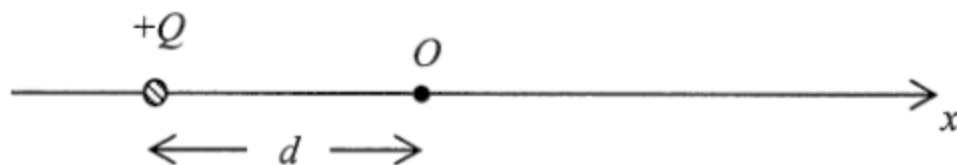


Which of the following statements **MUST BE** correct ?

- (1) Both spheres carry positive charges.
- (2) The amount of charges on the two spheres is the same.
- (3) The separation  $d$  depends on  $m$ .

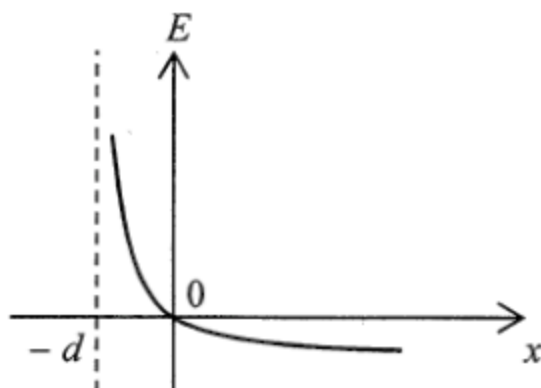
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

- \*23. A point charge  $+Q$  is fixed at a distance  $d$  away from the origin  $O$  as shown.

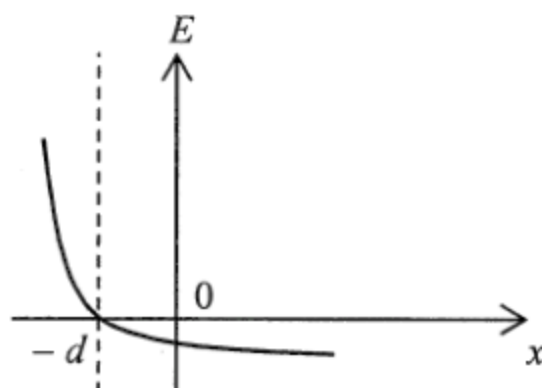


Which of the following graphs best represents the variation of the electric field strength  $E$  along the  $x$ -axis? (Take the electric field pointing to the right as positive.)

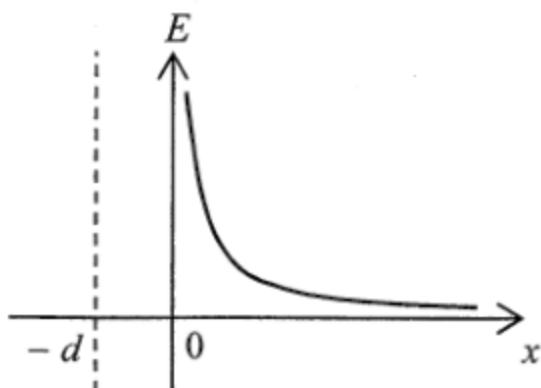
A.



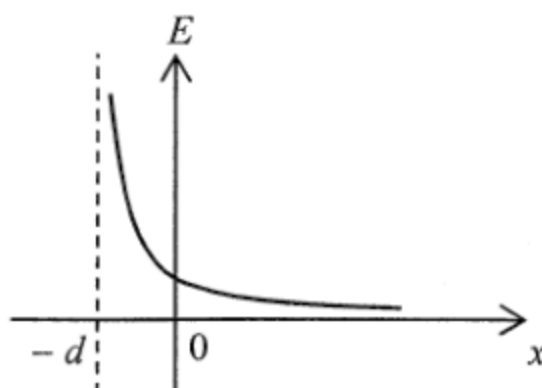
B.



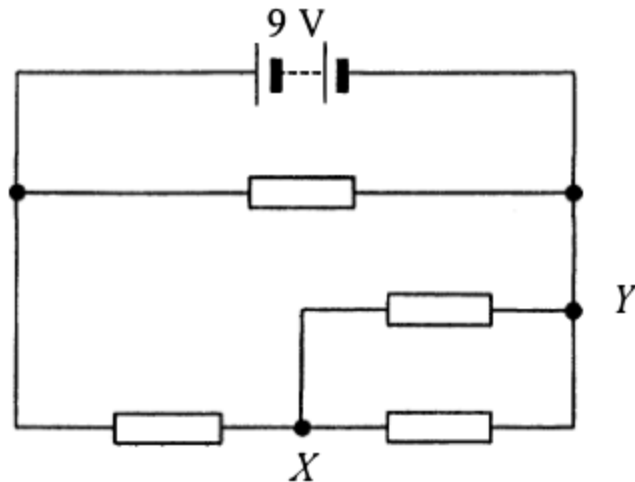
C.



D.



24. In the circuit, all resistors are identical. The internal resistance of the battery can be neglected.



What is the potential difference between  $X$  and  $Y$ ?

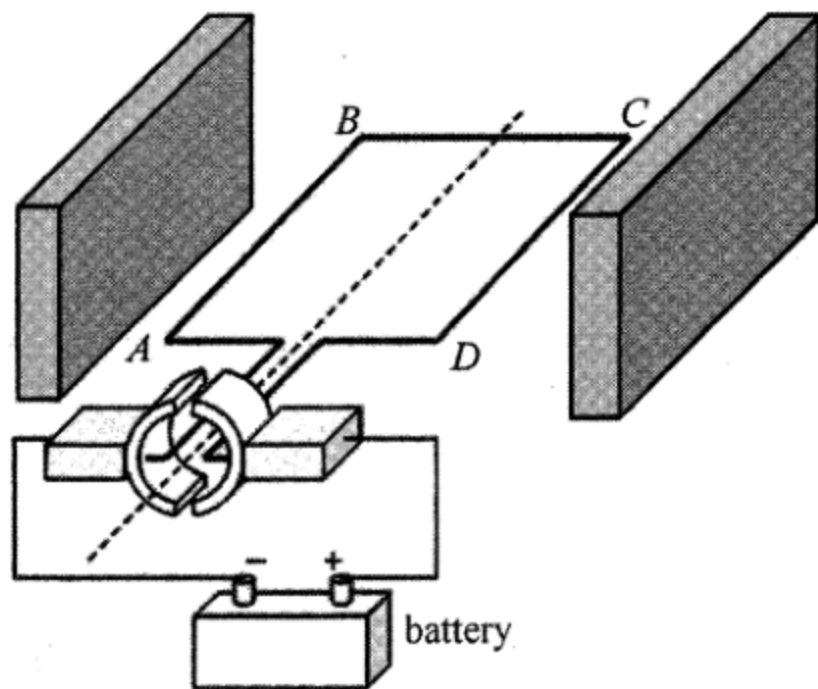
- A. 1.5 V
- B. 3.0 V
- C. 4.5 V
- D. 6.0 V



25. Which of the following statements about the use of a fuse is correct ?

- A. A fuse should be installed in the neutral wire.
- B. A fuse is not required in an electrical appliance with double insulation.
- C. A 5 A fuse is suitable for a heater of rating '220 V, 1500 W'.
- D. The melting point of a fuse should be lower than that of copper.

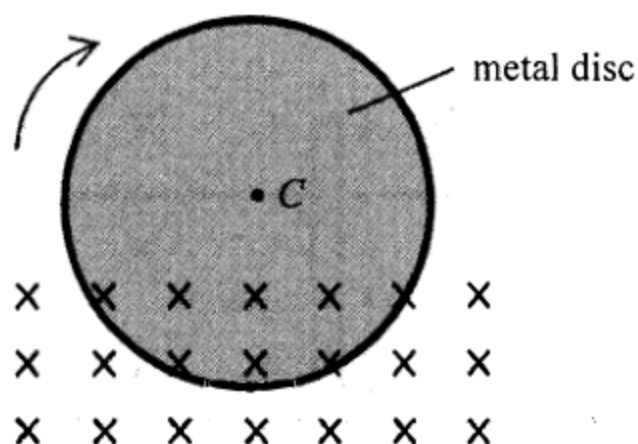
26. The figure shows a simple d.c. motor, the coil  $ABCD$  is mounted between the poles of two slab-sh magnets.



Which of the following statements is correct ?

- A. The turning effect is zero when the coil is vertical.
- B. The magnetic force acting on  $BC$  is the greatest when the coil is horizontal.
- C. The direction of the magnetic force acting on  $AB$  remains constant.
- D. The direction of the current in the coil remains unchanged.

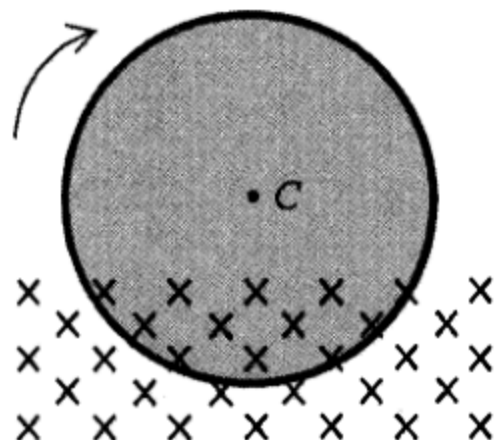
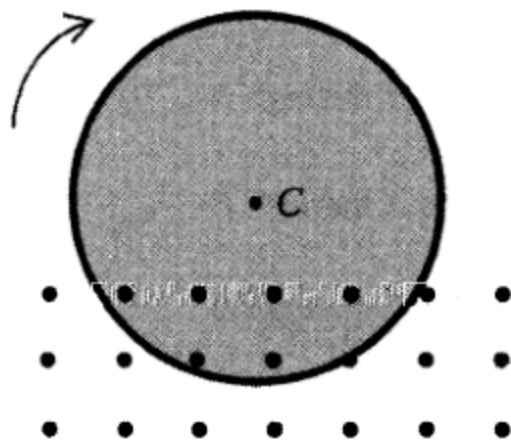
27. A metal disc is rotating about its centre  $C$  with constant speed. Part of the metal disc is in a uniform magnetic field pointing into the paper as shown. An eddy current flows in the metal disc.



After which of the following changes will the eddy current increase ?

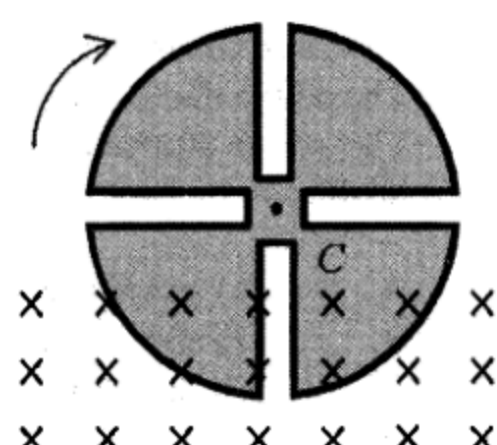
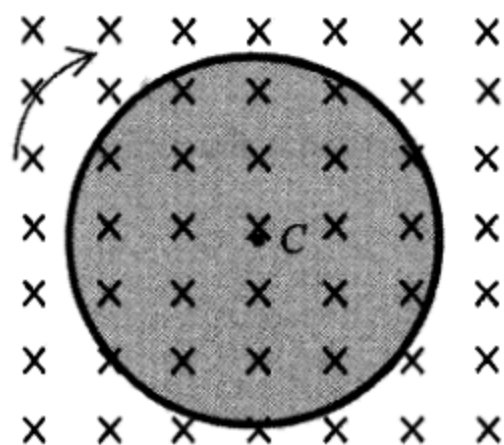
A. Reverse the direction of the magnetic field

B. Increase the strength of the magnetic field

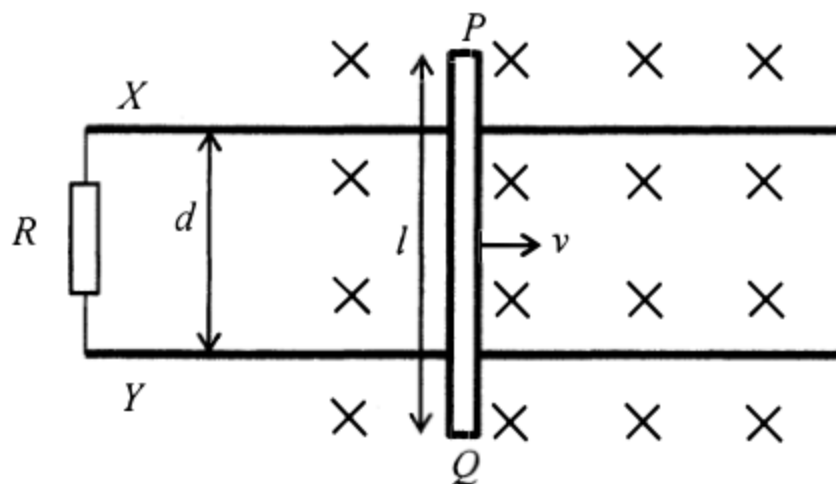


C. Apply the magnetic field over the whole metal disc

D. Cut several slits from the metal disc



- \*28. A metal rod  $PQ$  of length  $l$  is moving along smooth horizontal metal rails  $X$  and  $Y$  with constant speed  $v$  in a uniform magnetic field of magnetic field strength  $B$  pointing into the paper. The metal rails  $X$  and  $Y$  are separated by a distance of  $d$  and are connected to a resistor of resistance  $R$  as shown.



Which of the following descriptions about the induced current is correct ?

- |    | <b>magnitude</b> | <b>direction</b>            |
|----|------------------|-----------------------------|
| A. | $\frac{Blv}{R}$  | from $X$ to $Y$ through $R$ |
| B. | $\frac{Blv}{R}$  | from $Y$ to $X$ through $R$ |
| C. | $\frac{Bdv}{R}$  | from $X$ to $Y$ through $R$ |
| D. | $\frac{Bdv}{R}$  | from $Y$ to $X$ through $R$ |

\*29. A heater of resistance  $100\ \Omega$  is connected to the mains supply. The r.m.s. voltage of the mains supply is  $110\ \text{V}$ . Which of the following statements are correct ?

- (1) The peak voltage across the heater is  $156\ \text{V}$ .
- (2) The power dissipated by the heater is  $121\ \text{W}$ .
- (3) The power dissipated by the heater will be doubled if the r.m.s. voltage of the mains supply doubles.

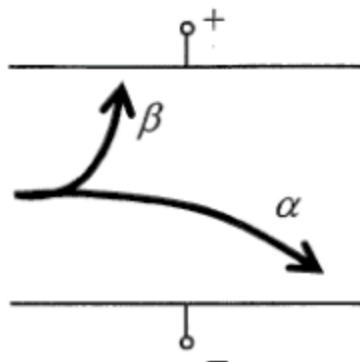
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

\*30. The input terminal of a transformer is connected to the 220 V mains supply. Ten identical light bulbs are connected in parallel to the output terminal of the transformer. All the light bulbs are working at their rated values of '3 V, 1.5 W'. If the efficiency of the transformer is 70%, what is the current drawn from the mains supply ?

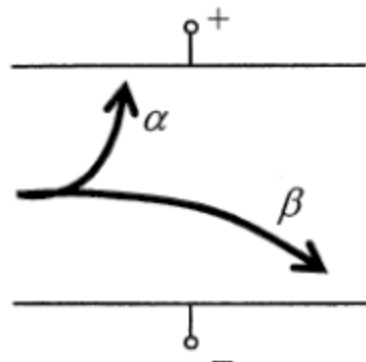
- A. 0.007 A
- B. 0.048 A
- C. 0.068 A
- D. 0.097 A

31. Which of the following diagrams best shows the deflection of  $\alpha$  and  $\beta$  particles in a uniform electric field in vacuum?

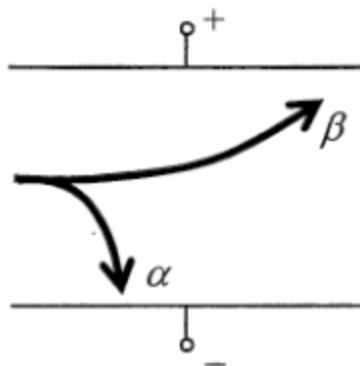
A.



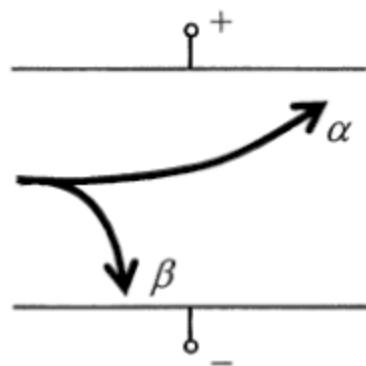
B.



C.



D.

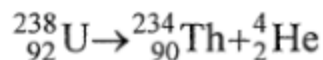


32. Which of the following statements about  $\beta$  particles and  $\gamma$  rays is correct ?

- A. Only  $\beta$  particles can ionize air particles.
- B. Only  $\gamma$  rays can travel through vacuum.
- C. Both of them can be detected by a photographic film.
- D. Both of them carry charge.



\*33. The following shows the decay of uranium-238 ( $^{238}_{92}\text{U}$ ).



Given that :      mass of  $^{238}_{92}\text{U} = 238.05079 \text{ u}$   
                         mass of  $^{234}_{90}\text{Th} = 234.04363 \text{ u}$   
                         mass of  $^4_2\text{He} = 4.00260 \text{ u}$

Which of the following statements is/are correct ?

- (1)      The temperature required to start the decay is about  $10^7 \text{ K}$ .
  - (2)      The energy released in the decay of one uranium-238 nucleus is  $4.25 \text{ MeV}$ .
  - (3)      All the energy released in the decay becomes the kinetic energy of  $^4_2\text{He}$ .
- 
- A.      (1) only
  - B.      (2) only
  - C.      (1) and (3) only
  - D.      (2) and (3) only

**END OF SECTION A**